

RIJAD ALISIC

Wallenberg Postdoctoral Fellow, Laboratory for Information and Decision Systems,
Massachusetts Institute of Technology

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EDUCATION

- 2018–2023 *Doctor of Philosophy & Licentiate of Engineering,*
KTH Royal Institute of Technology
Ph.D. thesis: Defense of Cyber-Physical Systems Using Learning-Based Attackers, 2023
Lic. thesis: Privacy of Sudden Events in Cyber-Physical Systems, 2021
Advisor: Prof. Henrik Sandberg, *Co-advisor:* Prof. Karl Henrik Johansson
- 2013–2018 *Master of Science in Engineering Physics,*
Faculty of Engineering LTH, Lund University
Grade average: 5/5
Specializations: Automatic Control and Theoretical Physics
Master thesis: Distributed Heating Networks, 2018
Advisor: Ass. Prof. Richard Pates, *Examiner:* Prof. Anders Rantzer

POSITIONS

- 2025–PRESENT *Wallenberg Postdoctoral Fellow,*
Laboratory for Information and Decision Systems, Massachusetts Institute of Technology
Advisor: Prof. Saurabh Amin
Affiliations: Lead Cybersecurity Researcher, MIT Maritime Consortium
- 2024–2025 *Postdoctoral Researcher, Wallenberg AI, Autonomous Systems and Software Program*
(WASP),
Division of Decision and Control Systems, KTH Royal Institute of Technology
Advisor: Prof. Henrik Sandberg
- 2016–2019 *Mathematician, Department of Experimental Medical Science, Lund University*
A Multi-object Tracker for Social Interactions of Zebrafish Larvae
- 2017 *Summer Undergraduate Research Fellow (SURF), California Institute of Technology*
A Controller Testbed using Temporal Logic, with Dr. Sumanth Dathathri and Prof. Richard Murray

PUBLICATIONS

Journal papers

- R. Alisic, J. Kim, and H. Sandberg, “Anomaly Detection with LWE Encrypted Control,” *IEEE Transactions on Automatic Control*, conditionally accepted, 2025. arXiv:2502.10283.
- R. Alisic, P. E. Paré, and H. Sandberg, “Change Time Estimation Uncertainty in Nonlinear Dynamical Systems with Applications to COVID-19,” *International Journal of Robust and Nonlinear Control*, 33(9):4732–4760, 2023.
- R. Alisic, J. Kim, and H. Sandberg, “Model-Free Undetectable Attacks on Linear Systems Using LWE-Based Encryption,” *IEEE Control Systems Letters*, vol. 7, pp. 1249–1254, 2023.

Conference papers

- J. Porzio, R. Alisic, and H. Sandberg, “Goodbye Innovation: Stealthy Destabilizing Sensor Attacks on Feedback Control Systems,” to be presented at the *65th IEEE Conference on Decision and Control (CDC)*, 2026.
- R. Alisic and S. Amin, “How Sensor Attacks Transfer Across Lie Groups,” to be presented at the *65th IEEE Conference on Decision and Control (CDC)*, 2026. arXiv:2604.03461.
- J. Porzio, A. Kanellopoulos, R. Alisic, G. Dán, and H. Sandberg, “SEC-Two: Secure Estimation over Two Channels,” *American Control Conference (ACC)*, 2026.
- R. Alisic and H. Sandberg, “Privacy Enhancement of Structured Inputs in Cyber-Physical Systems,” *60th IEEE Conference on Decision and Control (CDC)*, Austin, TX, 2021, pp. 4888–4894.

- R. Alisic and H. Sandberg, “Data-Injection Attacks Using Historical Inputs and Outputs,” *European Control Conference (ECC)*, Delft, 2021, pp. 1399–1405.
- R. Alisic, M. Molinari, P. E. Paré, and H. Sandberg, “Ensuring Privacy of Occupancy Changes in Smart Buildings,” *IEEE Conference on Control Technology and Applications (CCTA)*, Montreal, 2020, pp. 871–876.
- R. Alisic, M. Molinari, P. E. Paré, and H. Sandberg, “Maximizing Privacy in MIMO Cyber-Physical Systems Using the Chapman-Robbins Bound,” *59th IEEE Conference on Decision and Control (CDC)*, Jeju, 2020, pp. 6272–6277.
- R. Alisic, P. E. Paré, and H. Sandberg, “Modeling and Stability of Prosumer Heat Networks,” *IFAC-PapersOnLine*, vol. 52, no. 20, 2019, pp. 235–240.

AWARDS

2024 Top Downloaded Article 2022–2023, *International Journal of Robust and Nonlinear Control*, Wiley
 2020 Student Travel Award, *IEEE Control Systems Society*

AWARDED STIPENDS

2025 Knut and Alice Wallenberg Foundation, stipend for postdoctoral studies at MIT, Boston, MA
 2022 KTH Technical, Scientific, Research, and Education Fund, travel stipend for research exchange at the University of Melbourne
 2017 Summer Undergraduate Research Fellowship stipend, California Institute of Technology
 2017 Carl Erik Levin’s Fund; Academic Society’s Scholarship Fund; Sparbanken Syd’s Scholarship Fund
 2016 Archbishop H. Reuterdaahl’s Scholarship Fund; A.V. Bäcklund’s Scholarship Fund
 2016 Summer Scholarship for Undergraduate Research Project, Lund University
 2013 Torsten Tegstam’s Scholarship Fund, Sydsjænska Gymnasieförbundet

TEACHING AND SUPERVISION

2026 Supervisor, MSc Thesis, KTH Royal Institute of Technology
 “Cyber-Physical Security in Direct Data-Driven Control Systems,” Gustaf Zachrisson. Co-supervisors: Alf Isaksson (ABB), Srihalaji Coimbatore Anand (KTH). Examiner: Henrik Sandberg.
 2026 Supervisor, MSc Thesis, Massachusetts Institute of Technology
 “Anomaly Detection for GPS Spoofing in Maritime Cyber-Physical Systems: Filter-Based and Deep Learning Methods,” Strahinja Janjusevic. With Themistoklis Sapsis, Saurabh Amin, Fotini Christia.
 2025 Supervisor, MSc Thesis, KTH Royal Institute of Technology
 “Automatic PID Tuning for Wastewater Pump Station: Run Pumps Efficiently With the Press of a Button,” Balder Jörsäter. Co-advisor: Victor Molnö.
 2025 Supervisor, Summer Research Internship, Digital Futures, KTH
 “Synthetic Data Generation for Anomaly Detection,” Johan Slettengren. With Aris Kanellopoulos and Henrik Sandberg.
 2024 Supervisor, Summer Research Internship, KTH & University of California, Berkeley
 “Can Transformers Learn Kalman Filters?” Wentinn Liao. With Christos Mavridis, Henrik Sandberg, Anant Sahai.
 2021–2022 Teaching Assistant, Advanced Control Theory, KTH
 Offered twice.
 2021 Supervisor, Summer Research Internship, Digital Futures, KTH
 “Security of Smart Home Sensors,” Giorgio Sacchi.
 2021 Supervisor, MSc Thesis, KTH
 “Modeling and Evaluation of Turret Control Systems for Main Battle Tanks,” Mikael Lyth.
 2019 Supervisor, BSc Thesis, KTH
 “Distributed Optimisation in Multi-Agent Systems Through Deep Reinforcement Learning,” Andreas Eriksson & Jonas Hansson. Co-supervisor: Mina Ferizbegovic.
 2017–2020 Teaching Assistant, Basic Control Theory, KTH and Lund University
 Three times at each institution.

RESEARCH VISITS

2024 Seoul National University and Seoul National University of Science and Technology (SeoulTech)
 2024 Excellence Center at Linköping–Lund in Information Technology (ELLIIT), Lund University
 2022 University of Melbourne

INVITED TALKS

- 2026 *Invited Speaker, Workshop WS-08, IFAC World Congress, Busan, South Korea*
- 2025 *Invited Talk, Division of Systems and Control, Uppsala University*
- 2024 *Invited Speaker, ELLIIT Symposium, Lund University*
- 2024 *Invited Speaker, American Control Conference Student Workshop, Toronto*
- 2024 *Invited Lectures (two), SeoulTech, Seoul, South Korea*
- 2024 *Invited Seminar, Seoul National University, Seoul, South Korea*
- 2024 *Invited Talk, WASP Cybersecurity Cluster Meeting, Västervik*
- 2022 *Invited Seminar, Australian National University, Canberra*
- 2022 *Invited Seminar, University of Melbourne*

SERVICE

- FALL 2026–PRESENT *Chair, LIDS Postdoc Committee, MIT*
- 2025 *Panelist, WARA Operational Data Workshop, Ericsson, Stockholm*
- 2024 *Lead, Postdoctoral Workplace Meeting (arbetsplatsträff), Division of Decision and Control Systems, KTH*
- 2024 *Chair of Mechatronics Session, European Control Conference, Stockholm*
- 2024 *Volunteer and Election Assistant, European Control Association; Volunteer, European Control Conference, Stockholm*
- 2020–2022 *Lab Responsible, Division of Decision and Control Systems, KTH*
- 2020 *Organizer, KTH IFAC 2020 Workshop and NetCon Workshop on Learning, Stockholm*
- 2018 *Volunteer, Her Tech Future, Automatic Control, Lund*

PEER REVIEW

- JOURNALS *IEEE Transactions on Automatic Control; Automatica; IEEE Transactions on Control of Network Systems; IEEE Transactions on Cyber-Physical Systems; IEEE Control Systems Letters; IEEE Transactions on Industrial Informatics; IET Cyber-Physical Systems: Theory & Applications; IEEE Access*
- CONFERENCES *IEEE Conference on Decision and Control (CDC); American Control Conference (ACC); European Control Conference (ECC); IEEE Conference on Control Technology and Applications (CCTA); IFAC World Congress*

LANGUAGES AND SKILLS

- LANGUAGES: *Swedish and Bosnian (native), English (fluent), Spanish (basic)*
- ADVANCED: *Python, Matlab, $\LaTeX$*
- INTERMEDIATE: *Java, ROS*
- BASIC: *JavaScript, Maple*